

Linear Algebra (MT1004)

Final Exam

Date: June 4th 2024

Course Instructor:

Mr. Muhammad Jamil

Total Time (Hrs): 03

Total Marks: 50

Total Questions: 04

Roll No

Section

Student Signature

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Attempt all the questions.

CLO2 #: Understanding the core concepts of Euclidean vector spaces and matrix transformations

Q1

[05+05=10 Marks]

- a) For which real values of λ do the following vectors form a linearly dependent set in R^3 ?

$$v_1 = (\lambda, -\frac{1}{2}, -\frac{1}{2}), v_2 = (-\frac{1}{2}, \lambda, -\frac{1}{2}), v_3 = (-\frac{1}{2}, -\frac{1}{2}, \lambda)$$

- b) Consider the matrix $A = \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 0 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$

Find

- The basis for row & column space of A .
- The basis for null space of A .
- Verify dimension theorem of given matrix A .

CLO3 #: Applying the basic linear algebra concepts in computer science.

Q2

[05+05=10 Marks]

- a) Let M_{22} have the standard inner product defined by $U = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, V = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$

- Find the cosine of the angle between the vectors.
- Verify that $\langle u, v \rangle = \frac{1}{4} \|u + v\|^2 - \frac{1}{4} \|u - v\|^2$

- b) Express the matrix $A = QR$ decomposition, where Q matrix obtained by applying the Gram-Schmidt orthonormalization process to the column vector of A

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}, Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{6}} \end{bmatrix}$$

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CLO1 #: Interpreting and finding the solutions of linear equations in detail.

Q3

[05+10=15 Marks]

a) Find the standard matrix for the transformation $T: R^4 \rightarrow R^2$ defined by

$$w_1 = 2x_1 + 3x_2 - 5x_3 - x_4$$

$$w_2 = x_1 - 5x_2 + 2x_3 - 3x_4$$

and then compute $T(1, -1, 2, 4)$

b) Find a matrix P that diagonalizes $B = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ and determine B^9

CLO3 #: Applying the basic linear algebra concepts in computer science.

Q4

[05+10=15 Marks]

a) Determine whether the following matrix is orthogonal and if so find its inverse.

$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

b) Consider $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ and its eigenvectors $u_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

- i. Apply the Gram-Schmidt process to transform the basis, into an orthogonal basis.
- ii. Normalize the orthogonal basis vectors to obtain an orthonormal basis $Q = \{q_1, q_2, q_3\}$

ALL THE BEST